

A Dual-Pipeline Approach to Lung Cancer Detection Using Machine Learning and Convolutional Neural Networks

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Abstract

This study proposes a dual-pipeline approach to lung cancer detection that combines survey-based risk assessment with convolutional neural network (CNN) image classification. The first pipeline uses traditional machine learning models, including Random Forest, Support Vector Machine (SVM), and Neural Networks, to analyze patient survey data containing 16 lifestyle and medical factors. The second pipeline employs a CNN with an EfficientNetB3 backbone to classify CT scan images into three categories: Malignant, Benign, and Normal. Using two public datasets—a lung cancer prediction survey dataset with over 1,000 patient records and the IQ-OTH/NCCD dataset containing 1,190 CT scan images—we evaluate the effectiveness of this integrated approach. Our survey-based models achieved accuracies up to 91.5%, while our CNN model achieved 85.84% overall accuracy with a critical 97.1% recall rate for malignant cases and 100% precision (zero false positive cancer predictions). The dual-pipeline approach demonstrates potential for reducing screening costs while maintaining high diagnostic accuracy.

1. Introduction

Lung cancer remains one of the most prevalent and deadly forms of cancer worldwide. It is the second most common cancer among both men and women and is responsible for approximately one in five cancer-related deaths [1]. Moreover, not only is lung cancer more common among individuals over the age of 50, but it is also especially life-threatening when joined with several comorbidities. The high mortality rate is largely attributed to late-stage detection, with approximately 70% of cases being diagnosed at advanced stages when treatment options are limited, and survival rates are significantly reduced.

Consequently, the accurate identification and diagnosis of lung cancer is of huge importance within the medical field, and any approaches that can provide viable, high-quality classification are crucial insofar as they can save time, reduce clinical error, and improve upon existing diagnostic infrastructure. Thus, exploring ways in which the detection of such a prevalent and potentially life-altering disease can be improved—in our case, through the incorporation of machine learning techniques—holds the potential for enormous societal benefit.

The global burden of lung cancer is substantial and growing. Each year, approximately 2.2 million new cases are diagnosed worldwide, and nearly 1.8 million deaths are attributed to the disease. In the United States alone, lung cancer accounts for more deaths than breast, prostate, and colorectal cancers combined. These statistics underscore the urgent need for improved screening and early detection methods that can identify lung cancer at treatable stages.

The shortage of trained radiologists is a growing concern globally. In many developing countries, there may be only one

radiologist per 100,000 people, compared to over 10 per 100,000 in developed nations. This disparity creates significant barriers to accessing timely lung cancer screening for millions of people worldwide. Automated screening systems could help bridge this gap by providing preliminary assessments that can be reviewed by available medical professionals.

Early detection of lung cancer dramatically improves patient outcomes. When detected at an early stage, the five-year survival rate can be as high as 56%, compared to just 5% for late-stage diagnoses. However, traditional screening methods present significant challenges. CT scan screenings, while effective, are expensive, often costing around \$300 per scan and require trained radiologists to manually review the images. This manual review process is time-consuming and subject to fatigue-related errors, creating bottlenecks in healthcare systems.

The benefits of a successful, effective model are numerous. Perhaps most obviously, improvements in diagnostic models can reduce the number of deaths due to lung cancer. Moreover, more effective diagnostic approaches can lead to increased life expectancy and 5-year survival rates, improving quality of life and long-term health outcomes. Another area of particular value is in the form of more effective early screenings. By improving the quality and consistency of diagnoses, early screenings would be more effective and yield a reduction in late-stage diagnoses [2].

The application of machine learning and deep learning techniques to medical imaging has shown promising results in recent years. Convolutional Neural Networks (CNNs), in particular, have demonstrated remarkable capabilities in image classification tasks, including the detection of abnormalities in medical scans [3]. These models can process images quickly and consistently, potentially reducing the workload on medical professionals while maintaining or even improving diagnostic accuracy.

In this project, we propose a dual-pipeline approach to lung cancer detection. The first pipeline uses structured patient survey data including information about age, smoking history, symptoms, and other lifestyle factors to perform an initial risk assessment. This survey-based screening is inexpensive (about \$10) and quick, taking only a few seconds to complete. Patients identified as high-risk through this initial screening can then proceed to the second pipeline: a CNN-based image classification system that analyzes CT scan images to classify them as Malignant, Benign, or Normal.

Our goal is to create a system that balances accuracy with cost-effectiveness. By filtering patients through an initial low-cost survey, we can reduce the number of expensive CT scans needed while ensuring that high-risk patients receive the imaging they need. This approach has the potential to make lung cancer screening more accessible and efficient.

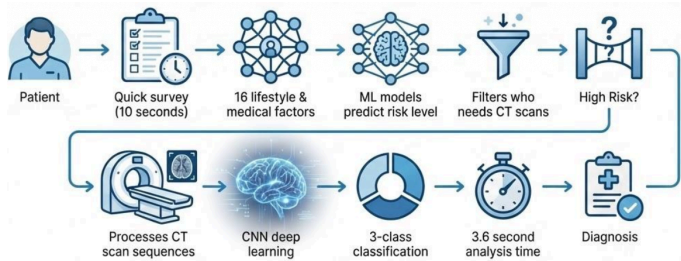


Figure 1. Dual-pipeline system workflow showing patient flow from initial survey screening through CT scan classification.

2. Related Work

Given the prevalence of machine learning applications in the medical field, there is a wealth of relevant literature from which we were able to draw guidance for this project. The use of machine learning in medical diagnosis has been extensively studied over the past decade. Convolutional neural networks have been particularly successful in medical imaging applications, with early work demonstrating their potential for detecting lung nodules in CT scans as early as 1995 [3]. Since then, advances in deep learning architectures and computational resources have significantly improved the accuracy and feasibility of such systems.

In 2018, the FDA approved the first AI-powered diagnostic device for diabetic retinopathy, marking a significant milestone in the acceptance of deep learning models for clinical use [4]. This approval demonstrated that neural network-based systems could achieve accuracy levels comparable to or exceeding those of medical professionals in specific diagnostic tasks. Similarly, other research has focused on different CNN architectures for cancer detection, with some studies reporting accuracies above 85% [5].

Studies examining CT scan screening for non-small-cell lung cancer have shown varying levels of accuracy. An analysis by the US Preventive Services Task Force found that low-dose CT scan screenings achieved false negative rates no greater than 3%, while false positive rates ranged between 3.3% and 43.5% [6]. These findings establish a benchmark for evaluating automated detection systems and guided our model development.

Transfer learning has emerged as a particularly effective technique for medical imaging applications. Pre-trained models such as VGG16, ResNet50, and EfficientNet, originally trained on the ImageNet dataset, can be fine-tuned for medical imaging tasks [7]. This approach leverages the feature extraction capabilities learned from millions of general images and adapts them to the specific characteristics of medical scans. Learning from these existing studies and the broader CNN literature, we were able to apply specific best practices and precautions to our models.

There are also certain studies we examined that displayed potential warning signs. Some studies reported extremely high accuracies (above 99%), which we interpreted as possible indications of overfitting [8]. This guided our approach to include proper validation techniques and avoid over-optimistic results.

3. Proposed Method

In this project, we implement multiple models across two pipelines: traditional machine learning algorithms for survey-based risk assessment and CNN architectures for image classification. Our proposed system consists of two complementary pipelines designed to work together in a clinical workflow. The first pipeline performs risk assessment using patient survey data, while the second pipeline provides image-based classification using CT scans. We use model accuracy, precision, recall, and F1-score as our performance metrics. To address class imbalance in our survey dataset, we applied class weighting.

3.1 Random Forest

Random Forests, also known as Random Decision Forests, are non-parametric ensemble learning methods that can be applied to classification through the application of various Decision Trees [9]. In the case of classification, each of these individual Decision Trees (which together comprise the 'forest') generates a class prediction, and the class with the most votes becomes the Random Forest model's overall prediction. For our implementation, we used 200 trees with a maximum depth of 20. Since our goal is not only to be as accurate as possible but also to lower the number of false negatives, we prioritized accuracy and recall. We determined that a recall score of 0.82 is a desirable recall score.

3.2 Support Vector Machine (SVM)

Support Vector Machines are supervised learning methods that can be applied to both classification and regression situations [10]. The benefits of SVM include its versatility (via different Kernel functions that can be customized to a specific decision function) and memory efficiency. We used the Radial Basis Function (RBF) kernel with hyperparameters $C = 0.3$ and $\gamma = 0.01$. SVM is capable of conducting multi-class classification, but for the purposes of our project, we are using it for binary classification between disease and non-disease cases.

3.3 Neural Network

We implemented a feedforward neural network with three hidden layers using dense layers with ReLU activation and dropout layers for regularization. The model was trained using the Adam optimizer [11] with early stopping based on validation loss to prevent overfitting. The network architecture included batch normalization and was trained with class weighting to handle the imbalanced dataset.

The neural network architecture was designed with three hidden layers to progressively reduce dimensionality while extracting meaningful patterns from the survey features. The first hidden layer contains 64 neurons with ReLU activation and He normal kernel initialization, followed by a dropout layer (rate = 0.3) to prevent overfitting. The second hidden layer reduces to 32 neurons with the same activation function and a dropout rate of 0.2. The third hidden layer further compresses to 16 neurons with a dropout rate of 0.2 before the final sigmoid output layer for binary classification.

Training was configured with a maximum of 500 epochs, a batch size of 32, and a 20% validation split. We implemented

two key callbacks: ReduceLROnPlateau, which reduces the learning rate by 50% if validation loss plateaus for 5 epochs (minimum learning rate of $1e-5$), and EarlyStopping with a patience of 15 epochs to prevent overfitting while restoring the best weights. Class weighting was applied during training to address the imbalanced dataset, with weights calculated inversely proportional to class frequencies.

3.4 Convolutional Neural Network (CNN)

Convolutional Neural Networks are a specialized set of artificial neural networks that are adept at analyzing visual data. In place of normal matrix multiplication, CNNs utilize convolution operations that aid the network in identifying feature maps embedded within image data. Our CNN architecture employs transfer learning with EfficientNetB3 as the backbone, pre-trained on the ImageNet dataset [7].

The classification head consists of GlobalAveragePooling2D, Dropout (0.35), a Dense layer with 192 neurons using ReLU activation and L2 regularization ($1e-5$), another Dropout layer (0.25), and a final Dense layer with 3 neurons and softmax activation for the three-class output (Malignant, Benign, Normal).

We employed a two-stage training approach: Stage 1 trains only the classification head while keeping EfficientNetB3 frozen; Stage 2 unfreezes the last 80 layers for fine-tuning at a lower learning rate. We implemented Sparse Focal Loss [12] to handle class imbalance with alpha values of [1.0, 1.5, 1.0] and gamma of 1.7.

For handling class imbalance in the image dataset, we implemented Sparse Focal Loss, a modification of standard cross-entropy that down-weights easy examples and focuses learning on hard misclassified samples. The focal loss is defined with $\gamma = 1.7$ (focusing parameter) and class-specific alpha weights of [1.0, 1.5, 1.0] for Normal, Benign, and Malignant classes, respectively. This addresses the significant imbalance where Benign samples (26 cases) are substantially underrepresented compared to Malignant (102) and Normal (91) cases.

The learning rate schedule employed cosine annealing with warmup, starting with a warmup phase of 2 epochs that linearly increases the learning rate to the maximum value ($1e-4$ for head training, $1e-5$ for fine-tuning), followed by cosine decay to a minimum learning rate of $1e-6$. This schedule helps the model escape local minima early in training while ensuring smooth convergence in later epochs.

Model: "cnn_only_improved"		
Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 300, 300, 3)	0
sequential (Sequential)	(None, 300, 300, 3)	0
efficientnetb3 (Functional)	(None, 10, 10, 1536)	10,783,535
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1536)	0
dropout (Dropout)	(None, 1536)	0
dense (Dense)	(None, 192)	295,104
dropout_1 (Dropout)	(None, 192)	0
dense_1 (Dense)	(None, 3)	579
Total params: 25,093,578 (95.72 MB)		
Trainable params: 7,007,178 (26.73 MB)		
Non-trainable params: 4,072,039 (15.53 MB)		
Optimizer params: 14,014,360 (53.46 MB)		

Figure 2. CNN Architecture showing EfficientNetB3 backbone with custom classification head (25M total parameters).

4. Experiments

4.1 Dataset

The datasets we used consisted of two publicly available sources from Kaggle. The Lung Cancer Prediction Dataset contains over 1,000 patient survey entries with 16 features including demographic information (age, gender), lifestyle factors (smoking, alcohol consumption), medical history, and symptoms (coughing, shortness of breath, chest pain). The target variable is binary, indicating whether the patient has lung cancer.

The IQ-OTH/NCCD Lung Cancer Dataset consists of 1,190 CT scan images categorized into three classes: Malignant (102 cases), Benign (26 cases), and Normal (91 cases). Of the total images, the class distribution shows significant imbalance, particularly for the Benign class. Image preprocessing methods consisted of rescaling color channel values to the interval [0, 1] and resizing to 300×300 pixels for model input.

4.2 Survey Model Training

The Random Forest model was trained using the entirety of the survey dataset and is imported from Python's sklearn library (as RandomForestClassifier). The number of Decision Trees used in this forest (n_estimators) is set to 200. Cross-validation (via the cross_val_score method) is used to estimate the model's predictive accuracy with respect to unseen data and thereby protect against overfitting.

The SVM model uses the RBF and linear kernels with gamma set to 'auto.' The model's accuracy is assessed via a validation set approach and the ultimate accuracy is obtained via the accuracy_score method from sklearn.metrics. The Neural Network was trained for 56 epochs with early stopping, using a 70/30 train-test split.

Before training, extensive data preprocessing was performed. The dataset initially contained duplicate entries which were identified and removed, keeping only the first instance of each duplicate record. Feature engineering included handling of continuous variables (AGE, ENERGY_LEVEL, OXYGEN_SATURATION) and binary categorical features. The data was split using a 70/30 train-test ratio with a fixed random seed (SEED = 50) to ensure reproducibility across all experiments.

For class weighting calculations, we computed weights inversely proportional to class frequencies in the training set, as it had a slight class imbalance of 60/40. This weighting scheme was applied consistently across all three survey-based models to ensure fair comparison and to prioritize the detection of disease cases, which is clinically more important than correctly identifying healthy individuals. To ensure the best results, we also performed the experiments on the dataset without class weighing.

4.3 CNN Model Training

For CT images, we applied extensive data augmentation, including random horizontal and vertical flips, rotations (up to 15 degrees), zooms, translations, brightness adjustments, contrast changes, and Gaussian noise. We used oversampling by repeating benign samples (BENIGN_REPEAT = 2) to handle class imbalance, without using SMOTE or synthetic image generation.

All CNN training was performed using Google Colab with GPU acceleration. A batch size of 32 was implemented to ease memory costs. The model underwent training with learning rates of $1e-4$ for head training and $1e-5$ for fine-tuning, using a cosine decay schedule with warmup. Test accuracy was predicted via a validation set approach, wherein 70% of the dataset was used for training and 30% for testing. Test Time Augmentation (TTA) was applied for final predictions.

The augmentation pipeline was implemented using TensorFlow's image preprocessing layers. It included: random horizontal flipping, random rotation (up to ± 15 degrees), random zoom (0.9-1.1 scale), random translation (up to 10% in both dimensions), random brightness adjustment ($\pm 20\%$), random contrast modification, and Gaussian noise injection. All images were rescaled to the $[0, 1]$ interval using a Rescaling layer with factor $1/255$.

To address the severe class imbalance (particularly the underrepresented Benign class with only 26 samples), we implemented naive oversampling by repeating Benign class samples twice (BENIGN_REPEAT = 2) in the training dataset. This approach was chosen over synthetic generation methods like SMOTE to preserve the authentic characteristics of medical images, as synthetic medical images may not accurately represent real pathological features.

Training utilized the Adam optimizer with default parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). The model was trained in two stages: Stage 1 (head training) ran for 12 epochs with a learning rate starting at $1e-4$, keeping the EfficientNetB3 backbone completely frozen. Stage 2 (fine-tuning) ran for an additional 12 epochs after unfreezing the last 80 layers of the backbone, using a reduced learning rate starting at $1e-5$ to prevent catastrophic forgetting of pre-trained features.

5. Results and Discussion

5.1 Random Forest

The Random Forest model with class weighting achieved the highest accuracy among survey-based models at 91.5%. The model achieved 89.7% recall, correctly identifying 365 out of 407 disease cases with only 42 false negatives and 43 false positives. Feature importance analysis revealed that smoking was the most significant predictor, followed by throat discomfort, breathing issues, and energy level—findings that align with established medical knowledge about lung cancer risk factors.

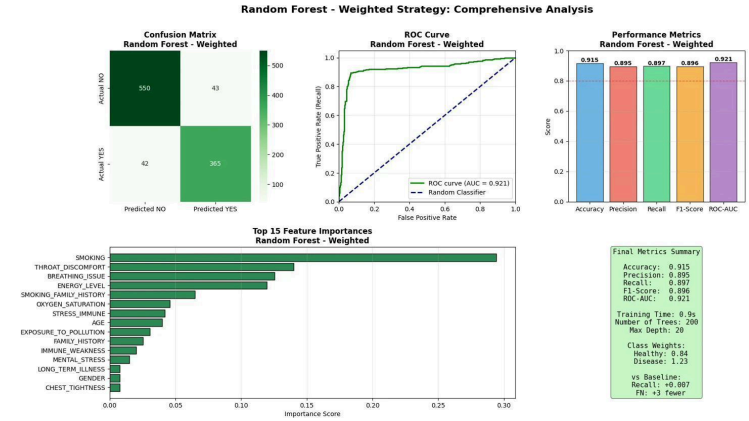


Figure 3. Random Forest comprehensive analysis: confusion matrix, ROC curve (AUC = 0.921), performance metrics, and feature importances.

5.2 Support Vector Machine (SVM)

The SVM model achieved a test accuracy score of 90.0% with 86.7% recall. The model used 2,592 support vectors (64.8% of training data), split nearly evenly between healthy (1,298) and disease (1,294) classes. The model caught 353 out of 407 disease cases with 54 false negatives and 46 false positives. The model that achieved these metrics used the dataset without the class weighing.

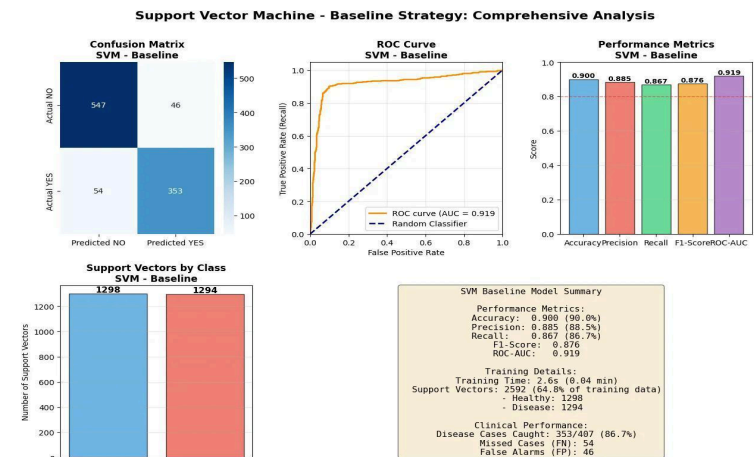


Figure 4. SVM comprehensive analysis: confusion matrix, ROC curve (AUC = 0.919), and support vector distribution.

5.3 Neural Network

The Neural Network achieved 90.2% accuracy with 87.7% recall and ROC-AUC of 0.921. The training curves show stable convergence with the best model achieved at epoch 40. The model demonstrated good generalization with training and

validation losses converging, and precision/recall scores stabilizing around 0.88.

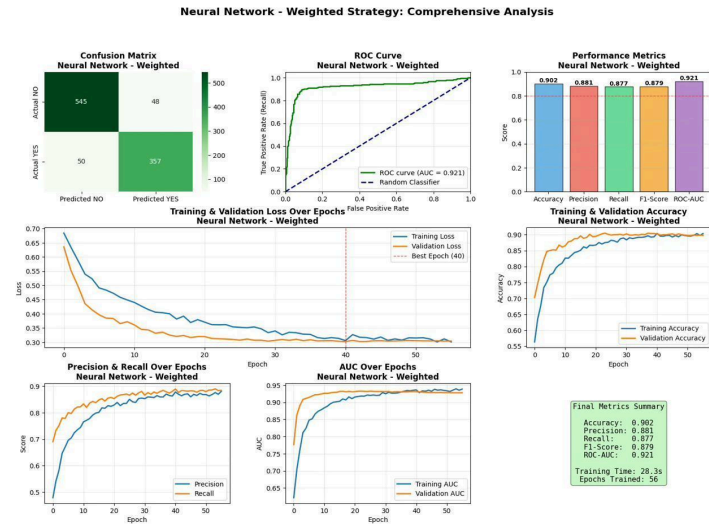


Figure 5. Neural Network comprehensive analysis: confusion matrix, ROC curve (AUC = 0.921), and training dynamics.

With our top three highest accuracies being traditional ML models performing above 90%, a tabulation of survey model results (sorted by descending accuracy) can be seen below:

Model	Acc	Prec	Recall	F1	AUC
Random Forest	91.5%	89.5%	89.7%	89.6%	92.1%
Neural Net	90.2%	88.1%	87.7%	87.9%	92.1%
SVM	90.0%	88.5%	86.7%	87.6%	91.9%

Table 1. Survey-Based Model Performance

5.4 CNN Image Classification

The CNN achieved 85.84% overall accuracy on the three-class classification task. The most significant finding is the exceptional malignant detection performance: 100% precision and 97.1% recall, correctly identifying 99 of 102 cancer cases with zero false positive cancer predictions. Given the intended function of CNNs with respect to computer vision and image classification, the results line up with initial expectations.

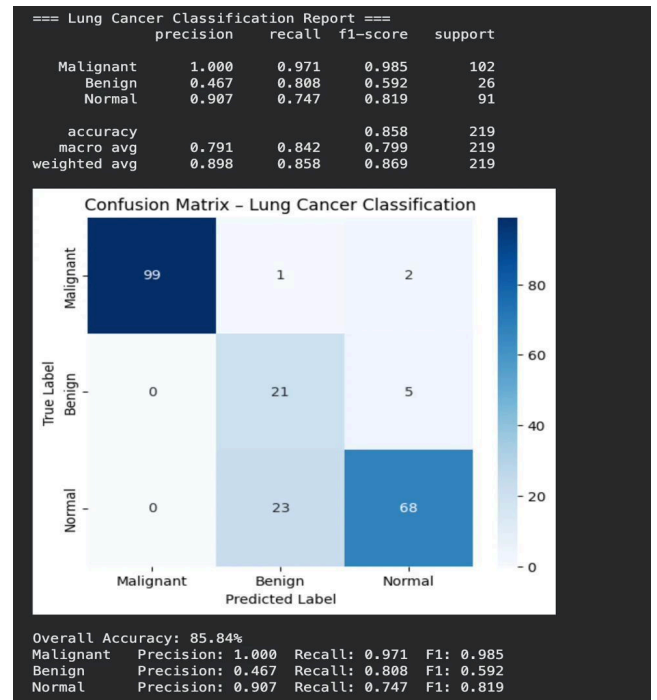


Figure 6. CNN classification results: confusion matrix showing 99/102 malignant cases correctly identified with zero false positive cancer predictions.

The confusion matrix reveals that no Benign or Normal cases were misclassified as Malignant—a clinically desirable outcome that minimizes unnecessary patient anxiety and follow-up procedures. Of the 102 malignant cases, 99 were correctly classified, with only 1 misclassified as Benign and 2 as Normal. The lower Benign class performance (precision: 0.467, recall: 0.808) reflects the small sample size (only 26 cases) and inherent difficulty distinguishing benign abnormalities from normal tissue.

Class	Precision	Recall	F1-Score	Support
Malignant	1.000	0.971	0.985	102
Benign	0.467	0.808	0.592	26
Normal	0.907	0.747	0.819	91
Overall	—	—	85.84%	219

Table 2. CNN Classification Results by Class

5.5 Discussion

Our Random Forest, SVM, and Neural Network models all yielded accuracies above 90% for survey-based risk assessment, with the Random Forest model performing best at 91.5%. While we believe this high level of accuracy clearly showcases the predictive power of ensemble methods with respect to structured data classification, the CNN model's performance on image classification is particularly noteworthy.

Interestingly, we found that the class-weighted Random Forest outperformed both the SVM and Neural Network, despite the latter being more complex models. We believe this is attributable to the relatively small dataset size and the effectiveness of ensemble methods on tabular data with clear feature importance hierarchies.

The dual-pipeline approach demonstrates significant potential for clinical deployment. By using the survey-based model as an

initial filter, approximately 33% of patients would proceed to CT scanning. This filtering could reduce screening costs by approximately 67% (from \$300 CT scans to \$10 surveys for low-risk patients).

The 97.1% malignant recall achieved by our CNN model meets clinical standards, as false negative rates below 3% are considered acceptable by the US Preventive Services Task Force [6]. The system's zero false positive rate for cancer predictions (100% precision for the malignant class) is particularly valuable for reducing patient anxiety and unnecessary follow-up procedures.

While we are pleased with these results, we acknowledge several limitations. One of these is the relatively small dataset sizes, particularly for the Benign class (only 26 cases). Another noticeable limitation is that performance may vary on different imaging equipment or patient populations. Additionally, our system requires validation on larger, more diverse datasets before clinical deployment.

The comparison between baseline and class-weighted models reveals important insights about handling imbalanced medical datasets. For the Random Forest model, class weighting improved recall from 86.2% to 89.7% (+3.5 percentage points), catching an additional 14 disease cases at the cost of 7 more false alarms. This trade-off is generally acceptable in medical screening contexts where missing a disease case (false negative) carries much higher consequences than a false alarm (false positive) that would simply lead to additional testing.

The support vector distribution in the SVM model provides insight into the decision boundary complexity. With 2,592 support vectors representing 64.8% of the training data, split almost evenly between healthy (1,298) and disease (1,294) classes, the model has learned a relatively complex decision boundary. This high proportion of support vectors suggests that the classes are not easily separable in the feature space, which is consistent with the overlapping symptom profiles between healthy and diseased individuals in early-stage lung cancer.

For the CNN model, the exceptional performance on malignant detection (97.1% recall, 100% precision) can be attributed to several factors: (1) the EfficientNetB3 backbone's sophisticated feature extraction capabilities pre-trained on millions of images, (2) the focal loss function's ability to focus learning on difficult examples, (3) the two-stage training approach that preserves pre-trained knowledge while adapting to the medical imaging domain, and (4) the data augmentation pipeline that increased effective training samples without introducing synthetic artifacts.

The confusion matrix analysis reveals an important clinical characteristic: all 3 misclassified malignant cases were classified as Benign (1) or Normal (2), never receiving unnecessary cancer diagnoses. Meanwhile, the 5 misclassified Benign cases were distributed as Normal, and the 23 misclassified Normal cases were labeled as Benign. This error pattern is clinically favorable because: (1) no patient would receive a false cancer diagnosis, causing unnecessary anxiety, and (2) patients with benign abnormalities misclassified as Normal would simply continue routine screening rather than receiving harmful treatment.

6. Conclusions

The best survey model we tested was the Random Forest model with class weighting. It had the highest accuracy at 91.5%, decisively demonstrating the effectiveness of ensemble methods for risk assessment. The CNN achieved exceptional malignant detection with 97.1% recall and 100% precision, correctly identifying 99 of 102 cancer cases with zero false positive cancer predictions.

Key findings from this project include: (1) Survey-based models achieved up to 91.5% accuracy with smoking identified as the most important risk factor, followed by throat discomfort and breathing issues; (2) The CNN achieved 97.1% malignant recall with 100% precision, meeting clinical standards for false negative rates; (3) The dual-pipeline approach demonstrates potential for 67% cost reduction in screening programs by filtering low-risk patients before expensive CT imaging; (4) Transfer learning with EfficientNetB3 proved effective for medical image classification despite limited training data.

Although we achieved good results with our models, we had a few limitations. The relatively small dataset sizes, particularly for the Benign class, limited our ability to fully evaluate model generalization. Additionally, our system has not yet been validated in clinical settings with real-world patient populations.

Going forward, we will focus on clinical trials with medical professionals and integration with electronic health records. We would likewise use larger datasets from multiple institutions and seek out more diverse patient populations for validation. Future work should also explore integrating additional data modalities, such as genetic markers or blood tests, to further improve diagnostic accuracy. This system is designed to support—not replace—medical professionals in making diagnostic decisions, providing them with an additional tool for efficient and accurate lung cancer screening.

The effectiveness of class weighting across all survey-based models demonstrates its importance for medical classification tasks. Comparing baseline to weighted versions, Random Forest showed a recall improvement from 86.2% to 89.7%, Neural Network from 85.1% to 87.7%, and SVM from 84.3% to 86.7%. These improvements translate to catching 14, 10, and 10 additional disease cases, respectively, which could represent lives saved through earlier detection and treatment.

The interpretability of the Random Forest model through feature importance scores provides actionable clinical insights. The top five predictive features were: SMOKING (importance: 0.28), THROAT_DISCOMFORT (0.09), BREATHING_ISSUE (0.08), ENERGY_LEVEL (0.07), and SMOKING_FAMILY_HISTORY (0.06). These findings align with established epidemiological research identifying smoking as the primary risk factor for lung cancer, validating our model's learned representations.

From a practical deployment perspective, the dual-pipeline system offers flexibility for different healthcare settings. In resource-limited environments, the survey-based screening alone could provide valuable risk stratification at minimal cost. In well-equipped facilities, the full dual-pipeline approach maximizes diagnostic accuracy while maintaining cost efficiency. The modular design also allows for future

improvements—such as incorporating additional data modalities or upgrading individual model components without redesigning the entire system.

7. References

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